



CYBER SECURITY ANALYST INTERNSHIP ENGAGEMENT REPORT

Week 01: Cybersecurity Fundamentals & Security Concepts

ORGANIZATION	IT SIMPLERA
REPORT TITLE	Week 01: Cybersecurity Fundamentals & Security Concepts
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1. Executive Summary

Cybersecurity is the practice of protecting computer systems, networks, applications, and digital information from cyber threats and unauthorized access. As technology becomes a part of everyday life, cyberattacks are increasing in both number and complexity. Individuals, businesses, and government organizations rely on cybersecurity to protect sensitive information, maintain business operations, and ensure the privacy of users.

A strong understanding of cybersecurity fundamentals is essential for identifying threats, preventing attacks, and responding effectively to security incidents. This report explores important cybersecurity concepts, including the CIA Triad, real-world cyberattacks, commonly used cybersecurity tools, secure home network design, and the daily responsibilities of a Cybersecurity Analyst. The activities completed in this report provide both theoretical knowledge and practical understanding of modern cybersecurity practices.

Task Completion Status

Task	Description	Status
Task 01	Research five real-world cyberattacks	Completed
Task 02	Identify which CIA Triad component	Completed
Task 03	Research three cybersecurity tools	Completed
Task 04	Create a secure home network diagram	Completed
Task 05	What Does a Cybersecurity Analyst Do Every Day	Completed

2. Learning Objectives

The primary objective of this assignment is to develop a comprehensive understanding of fundamental cybersecurity concepts and their practical applications in today's digital world. As technology continues to evolve, cyber threats have become more frequent and sophisticated, making cybersecurity knowledge essential for individuals and organizations alike. This assignment is designed to strengthen both theoretical knowledge and practical awareness by exploring important security principles, analyzing real-world cyber incidents, studying commonly used cybersecurity tools, and understanding the responsibilities of cybersecurity professionals. It also aims to improve research, analytical thinking, technical documentation, and report-writing skills, which are essential for a successful career in the cybersecurity field.

The specific learning objectives of this assignment are as follows:

- **To understand the basic concepts of cybersecurity and information security.**

Develop a clear understanding of cybersecurity fundamentals, including the importance of protecting computer systems, networks, applications, and sensitive information from cyber threats and unauthorized access.

- **To learn the CIA Triad (Confidentiality, Integrity, and Availability) and its importance.**

Understand the three core principles of information security and learn how they are applied to protect data, maintain system reliability, and ensure secure communication in real-world environments.

- **To research and analyze real-world cyberattacks and their impact.**

Study well-known cyberattacks to understand how they occurred, the vulnerabilities that were exploited, their consequences, and the lessons organizations learned from these incidents.

- **To identify commonly used cybersecurity tools and understand their functions.**

Explore widely used cybersecurity tools, their primary purposes, key features, and practical applications in network monitoring, vulnerability assessment, and security testing.

- **To design a secure home network using essential security components.**

Learn how to create a secure home network by incorporating security measures such as routers, firewalls, antivirus software, strong authentication, and optional VPN protection to reduce cyber risks.

- **To understand the daily roles and responsibilities of a Cybersecurity Analyst.**

Gain insight into the routine tasks performed by cybersecurity professionals, including threat monitoring, incident response, vulnerability management, security analysis, and risk assessment.

- **To improve technical research, documentation, and report-writing skills.**

Develop the ability to collect information from reliable sources, organize technical content professionally, and present cybersecurity findings in a structured and well-documented report.

- **To develop awareness of cybersecurity best practices for protecting systems and data.**

Understand the importance of implementing security best practices such as regular software updates, strong password policies, multi-factor authentication, data backup, and continuous security monitoring to protect digital assets.

3. Task Documentation

3.1 Task 01 — Research on five real-world cyberattacks

Cyberattacks have affected organizations worldwide, causing financial losses, operational disruption, and data breaches. Studying these incidents helps security professionals understand common attack methods and develop stronger defense strategies.

Attack	Year	Organization	Type	Description	Impact
WannaCry	2017	NHS, FedEx, Renault	Ransomware	WannaCry spread rapidly by exploiting a Windows vulnerability. It encrypted files and demanded a ransom payment to restore access.	More than 230,000 computers in over 150 countries were infected, disrupting hospitals, businesses, and government services.
Equifax Data Breach	2017	Equifax	Data Breach	Attackers exploited a vulnerability in a web application to steal personal information of customers.	Sensitive information of approximately 147 million people was exposed, leading to financial losses and legal actions.
SolarWinds Attack	2020	SolarWinds	Supply Chain Attack	Attackers inserted malicious code into official software updates, allowing them to access multiple organizations through trusted software.	Several government agencies and private companies were compromised, making it one of the largest supply chain attacks.
Colonial Pipeline Attack	2021	Colonial Pipeline	Ransomware	A ransomware attack forced the company to shut down fuel pipeline operations temporarily.	Fuel distribution was disrupted across several U.S. states, causing shortages and economic impact.

Attack	Year	Organization	Type	Description	Impact
Yahoo Data Breach	2013	Yahoo	Data Breach	Cybercriminals gained unauthorized access to Yahoo user accounts and stole account information.	Nearly 3 billion user accounts were affected, making it one of the largest data breaches in history.

3.2 Task 02 — CIA Triad Analysis

The CIA Triad is a fundamental cybersecurity model that focuses on protecting information through **Confidentiality, Integrity, and Availability**. Every cyberattack affects one or more of these security principles. Understanding which component is compromised helps organizations improve their security controls and reduce future risks.



Cyberattack	CIA Component	Explanation
WannaCry	Availability	The ransomware encrypted files and prevented users from accessing their systems until a ransom was paid.
Equifax Breach	Confidentiality	Attackers stole confidential customer information, including personal and financial data.
SolarWinds Attack	Integrity	The attackers modified trusted software updates, compromising the integrity of the software supply chain.
Colonial Pipeline	Availability	Business operations were interrupted, making fuel services temporarily unavailable.
Yahoo Data Breach	Confidentiality	Username, passwords, and personal information were exposed without authorization.

3.3 Task 03 — Research on Cybersecurity Tools

Cybersecurity professionals use specialized tools to monitor networks, identify vulnerabilities, and investigate security incidents. These tools improve the efficiency of security operations and help organizations detect potential threats before they cause damage.

Tool	Purpose	Function	Common Use Case
Wireshark	Network Analysis	Captures and analyzes network traffic in real time to identify communication between devices.	Network troubleshooting, packet analysis, and incident investigation.
Nmap (Network Mapper)	Network Scanning	Discovers live hosts, open ports, operating systems, and running services on a network.	Network auditing, vulnerability assessment, and penetration testing.
Burp Suite	Web Application Security Testing	Intercepts and analyzes web requests to identify vulnerabilities in web applications.	Web penetration testing, vulnerability assessment, and application security testing.

3.4 Task 04 — Secure Home Network Design

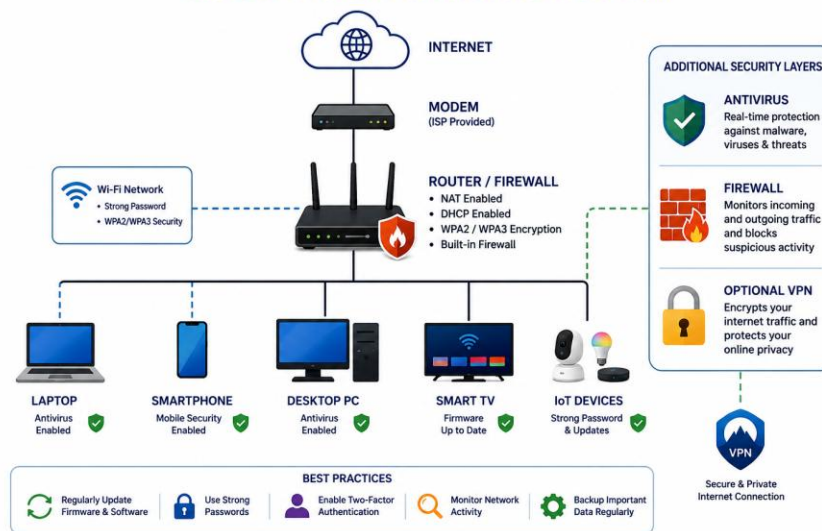
Introduction

A secure home network is essential for protecting personal devices, sensitive information, and internet communications from cyber threats. With the increasing use of laptops, smartphones, smart TVs, and IoT devices, home networks have become attractive targets for hackers. Weak passwords, outdated software, and unsecured Wi-Fi networks can expose users to malware, phishing attacks, and unauthorized access.

To reduce these risks, a secure home network should include multiple layers of security such as a modem, a router with firewall protection, antivirus software, regular software updates, and strong Wi-Fi encryption. Additional security measures, including a Virtual Private Network (VPN), can further protect online activities by encrypting internet traffic. A properly designed home network helps maintain the confidentiality, integrity, and availability of personal data.

Secure Home Network Diagram

SECURE HOME NETWORK DIAGRAM



Network Security Components

Component	Purpose
Internet	Provides connectivity to online services and websites.
Modem	Connects the home network to the Internet Service Provider (ISP).
Router	Distributes internet access to all connected devices.
Firewall	Filters incoming and outgoing traffic to block unauthorized access.
Antivirus	Detects and removes malware, viruses, and malicious software.
VPN	Encrypts internet traffic to improve online privacy and security.

3.5 Task 05 — What Does a Cybersecurity Analyst Do Every Day?

Introduction

A Cybersecurity Analyst is responsible for protecting an organization's digital systems, networks, and sensitive information from cyber threats. They continuously monitor security systems, identify suspicious activities, investigate security incidents, and implement security measures to reduce organizational risks. Their work plays an important role in maintaining business continuity and protecting valuable information.

The daily responsibilities of a Cybersecurity Analyst require technical knowledge, problem-solving skills, and continuous learning because cyber threats evolve rapidly. Analysts work closely with IT teams to strengthen security controls, respond to incidents, and ensure compliance with organizational security policies.

Daily Responsibilities of a Cybersecurity Analyst

A Cybersecurity Analyst performs several important tasks every day to maintain the security of an organization's infrastructure. These responsibilities include:

- Monitoring networks and security alerts for suspicious activities.
- Investigating security incidents and responding to cyber threats.
- Performing vulnerability assessments and recommending security improvements.
- Installing security patches and software updates.
- Managing firewalls, antivirus software, and intrusion detection systems.
- Reviewing system logs to identify unusual behavior.
- Conducting security awareness sessions for employees.
- Preparing security reports and documenting incidents.
- Ensuring compliance with cybersecurity policies and industry standards.
- Researching new cyber threats and recommending protective measures.

These responsibilities help organizations detect threats early, minimize security risks, and maintain the confidentiality, integrity, and availability of information.

4. Challenges Faced

During the completion of this assignment, several technical and research-related challenges were encountered. Understanding cybersecurity concepts required careful study because different attacks affect different security principles. Selecting appropriate cybersecurity tools and analyzing their functions also required additional research from reliable sources.

Another challenge was designing a secure home network that follows cybersecurity best practices while remaining simple and practical. Despite these difficulties, the assignment significantly improved my understanding of cybersecurity fundamentals, network security, and professional documentation.

5. Conclusion

This assignment provided a comprehensive understanding of essential cybersecurity concepts and their practical applications. Through the study of real-world cyberattacks, the CIA Triad, cybersecurity tools, secure network design, and the responsibilities of a Cybersecurity Analyst, valuable knowledge was gained regarding modern cybersecurity practices.

The activities completed during this assignment strengthened both theoretical understanding and practical awareness of information security. Overall, this report

highlights the importance of cybersecurity in protecting digital assets and demonstrates how proper security measures can help organizations and individuals defend against evolving cyber threats.

6. References

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